



ZAMREN HACKATHON 2026

SOFTWARE REQUIREMENTS SPECIFICATION

Zambia Digital Identity System

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Thematic Area:	Zambia Digital ID
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Zambia Digital Identity (ZDID) System. The system aims to modernize Zambia's national identity infrastructure by creating a unified, secure, and interoperable digital platform that serves as the backbone for government services, financial inclusion, and citizen empowerment. This document is intended for the development team, project stakeholders, judges, and mentors involved in the 2026 ZAMREN Student Hackathon.

1.2 Scope

The Zambia Digital ID System encompasses three integrated sub-systems:

- Government Administrative Dashboard (Control Center)
- Hospital Registration Application (Civil Registration System)
- Citizen Mobile Application (Digital ID Wallet)

Together, these sub-systems address the full citizen lifecycle - from birth registration and national ID enrollment through identity verification and third-party KYC integration via the Government Service Bus (GSB).

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
ZDID	Zambia Digital Identity: the overarching system being developed
DIN	Digital Identity Number: unique 12-character identifier assigned to each registered citizen

Term	Definition
NRC	National Registration Card: Zambia's existing physical identity card
GSB	Government Service Bus: the middleware integration layer connecting government systems
KYC	Know Your Customer: process of verifying the identity of a citizen or client
RO	Registration Officer: government official responsible for verifying citizen registrations
ICD-11	International Classification of Diseases, 11th Revision: used for death cause coding
HSM	Hardware Security Module: tamper-resistant hardware for cryptographic key management
OIDC	OpenID Connect: identity layer built on top of OAuth 2.0
NFC	Near Field Communication: short-range wireless connectivity for physical card interaction
AES-256	Advanced Encryption Standard with 256-bit key: encryption standard for data at rest
ECDSA	Elliptic Curve Digital Signature Algorithm: used for digital signatures
SRS	Software Requirements Specification: this document
ZAMREN	Zambia Research and Education Network

1.4 Overview

The remainder of this document is organized as follows: Section 2 provides an overall system description and context. Section 3 details all functional requirements organized by system interface. Section 4 specifies non-functional and security requirements. Section 5 presents the system design through UML diagrams. Section 6 covers technical architecture and deployment. Section 7 describes data requirements, and Section 8 outlines security and compliance standards.

2. Overall Description

2.1 Product Perspective

The Zambia Digital ID System is a new, integrated platform designed to modernize the fragmented, paper-based identity management processes currently in use across Zambia. The system interfaces with:

- The existing National Registration system (via the GSB)
- Mobile money platforms (MTN MoMo, Airtel Money, Zamtel Money)
- Third-party financial institutions and utility providers
- Healthcare facilities and field health workers
- Citizens via mobile devices (Android and iOS)

2.2 Product Context and Background

Zambia currently faces several critical challenges in identity management:

- Fragmented databases spread across multiple government departments with no interoperability
- Manual, paper-based registration processes that cause delays, errors, and opportunities for fraud

- Limited coverage in rural and remote areas where citizens lack access to registration services
- Security vulnerabilities in legacy systems susceptible to identity theft and fraud
- No unified digital channel for citizens to manage and share their identity data

The ZDID System directly addresses these challenges by providing a unified, cryptographically secure, and citizen-controlled digital identity ecosystem that complies with the Zambia Data Protection Act.

2.3 User Classes and Characteristics

The system serves the following distinct user classes:

2.3.1 Citizens

Any Zambian individual who holds or is eligible for a National Registration Card. Citizens interact with the system through the Citizen Mobile Application to enroll, manage, and share their digital identity. Varying levels of digital literacy must be accommodated, requiring a multilingual, intuitive interface.

2.3.2 Healthcare Professionals (Nurses and Doctors)

Qualified medical personnel at hospitals, clinics, and remote health posts. They are responsible for recording vital events (births and deaths) through the Hospital Registration App. Field health workers may operate in low-connectivity environments, requiring offline-first capabilities.

2.3.3 Registration Officers

Government officials with basic computer skills and knowledge of government registration procedures. They are responsible for verifying and approving digital ID registrations for citizens through the Administrative Dashboard.

2.3.4 Registrars

Senior officials with both technical and administrative expertise. Registrars manage the enrollment of Registration Officers and healthcare professionals, and vet third-party institutions requesting access to the GSB.

2.3.5 Data Analysts / Supervisors

Technically and analytically skilled individuals responsible for monitoring system activity, reviewing immutable audit trails, and generating national statistics on registration trends. They ensure system integrity and regulatory compliance.

2.3.6 Third-Party Institutions

External entities such as banks, utility companies, and financial service providers that have been vetted and approved to request citizen KYC information via the GSB, subject to explicit citizen consent.

2.4 Operating Environment

- Backend: Cloud-native microservices deployed via Docker and Kubernetes
- Frontend (Dashboard): Web application accessible via modern browsers
- Mobile (Citizen App): React Native supporting Android and iOS
- Mobile (Hospital App): Android APK with offline-first capability for rural health workers
- Database: PostgreSQL for relational data, with Redis for caching and message queuing via RabbitMQ
- Communication: All external communication over HTTPS/TLS 1.3 and internal services communicates via REST APIs

2.5 Design and Implementation Constraints

- All solutions must comply with the Zambia Data Protection Act
- Security standards must meet or exceed those specified in Section 4.2 of this document
- The system must support a minimum of seven Zambian languages: English, Bemba, Nyanja, Tonga, Lozi, Kaonde, and Lunda
- The Citizen App must function in offline mode for identity verification

- All APIs must follow RESTful conventions with OpenAPI 3.0 documentation
- Biometric data must comply with ISO/IEC 19794 series standards
- Date formats must conform to ISO 8601; country codes to ISO 3166-1

3. Functional Requirements

3.1 Citizen Mobile Application

3.1.1 Digital Enrollment and Registration

The Citizen App shall enable a hybrid enrollment process where citizens initiate their registration digitally before completing biometric verification in person.

ID	Requirement	Description	Priority
FR-C01	NRC Entry and Key Generation	The app shall prompt the citizen to enter their NRC number, full name, and phone number. Upon submission, the app shall generate an asymmetric cryptographic key pair (RSA-2048 or ECDSA P-256). The private key shall be stored in the device hardware security module or secure enclave. The public key shall be transmitted to the backend and stored in the pending registration table.	High
FR-C02	Pending Status Display	The app shall display a pending digital ID template with a 'Visit Office' notice after initial enrollment. The digital signature shall be inactive for official services but active for registration tracking.	High

ID	Requirement	Description	Priority
FR-C03	Activation via Biometric Challenge	Upon Registration Officer approval, the app shall receive an activation challenge from the server. The citizen shall authenticate using biometrics (fingerprint or Face ID) to unlock the private key, sign the challenge, and return it to the server for verification.	High
FR-C04	Digital ID Download and Storage	Once the government approves a registration, the citizen shall use the app to claim and download their digital ID. The ID shall be encrypted at rest using AES-256 and stored locally on the device.	High

3.1.2 Identity Verification and Sharing

ID	Requirement	Description	Priority
FR-C05	Dynamic QR Code Generation	The app shall generate a unique, signed QR code each time the citizen needs to display their identity. The QR code shall encode the citizen's DIN and expiry information, signed with the citizen's private key.	High
FR-C06	NFC Integration	The app shall support NFC-based identity verification, allowing the citizen's device to interact with physical ID card readers.	High

ID	Requirement	Description	Priority
FR-C07	Offline Verification	The app shall be capable of verifying the citizen's identity and displaying the digital ID without an active internet connection using locally stored cryptographic proofs.	High
FR-C08	Secure Login	Citizens shall authenticate using their device biometrics to sign a dynamic challenge for login into the app and third-party services via OAuth 2.0 / OpenID Connect.	High

3.1.3 KYC Consent Management

ID	Requirement	Description	Priority
FR-C09	KYC Request Notification	The app shall notify the citizen when a third-party institution requests access to their KYC information via the GSB.	High
FR-C10	Selective Data Sharing	The citizen shall be able to view the specific information being requested and selectively approve or deny access to individual data fields before sharing.	High
FR-C11	Consent Audit Trail	All consent approvals and denials shall be logged with timestamps in the immutable audit trail.	High

ID	Requirement	Description	Priority
FR-C12	Transaction Authorization	The citizen shall be able to authorize financial transactions from third-party institutions via the digital wallet, authenticated through biometrics.	Medium

3.1.4 Family Tree and Accessibility

ID	Requirement	Description	Priority
FR-C13	Family Tree View	Citizens shall be able to view their family lineage tree, populated automatically from birth and death records linked during registration.	Medium
FR-C14	Multi-language Support	The app shall support English, Bemba, Nyanja, Tonga, Lozi, Kaonde, and Lunda, selectable by the user at any time.	High
FR-C15	Cross-App Communication via GSB	The Citizen App shall interact with the Government Service Bus to facilitate KYC data sharing, strictly only upon explicit citizen consent.	High

3.2 Hospital Registration Application

3.2.1 Birth Registration

ID	Requirement	Description	Priority
FR-H01	Hospital Birth Registration	Health workers shall be able to digitally record a newborn's birth and link it to the mother's 12-character DIN. The record shall automatically trigger the issuance of a digital birth certificate.	High
FR-H02	Home Birth Support	The app shall support recording of births that occur outside of hospital facilities, with all mandatory fields accessible in offline mode.	High
FR-H03	Digital Birth Certificate Generation	Upon registration approval, the system shall generate a verifiable digital birth certificate. A printable version shall be available for citizens without smartphones.	High

3.2.2 Death Registration

ID	Requirement	Description	Priority
FR-H04	Death Record with ICD-11 Coding	Health workers shall record death events using ICD-11 (International Classification of Diseases, 11th Revision) cause-of-death classification codes to ensure accurate national health statistics.	High
FR-H05	Identity Retirement	Upon approved death registration, the deceased citizen's DIN shall be securely retired in the registry, preventing future use while preserving the historical record.	High

ID	Requirement	Description	Priority
FR-H06	Death Certificate Generation	The system shall generate a verifiable digital death certificate linked to the retired DIN.	High

3.2.3 Offline and Accessibility Features

ID	Requirement	Description	Priority
FR-H07	Offline-First Data Entry	The Hospital App shall support full data entry without an active internet connection. Records shall be queued and synchronized to the server once connectivity is restored.	High
FR-H08	Mobile Registration for Rural Workers	A dedicated mobile version of the registration form shall be available and optimized for field health workers in remote areas.	High
FR-H09	Multi-language Support	The app shall support all seven Zambian languages specified in Section 2.5.	High
FR-H10	Statistics Dashboard	The app shall provide a reporting dashboard showing birth and death statistics by region and time period for Supervisors.	Medium

3.3 Government Administrative Dashboard

3.3.1 Registration Officer Functions

ID	Requirement	Description	Priority
FR-A01	Citizen Registration Review	Registration Officers shall be able to view and review pending citizen registration requests submitted through the Citizen App.	High

ID	Requirement	Description	Priority
FR-A02	Physical Document Verification	The dashboard shall enable ROs to verify the citizen's physical NRC against the legacy National Registration system via the GSB.	High
FR-A03	Biometric Capture	The dashboard shall support capture of citizen biometric data (fingerprint and facial recognition) during in-person registration to prevent use of stolen NRCs or deepfakes.	High
FR-A04	Registration Approval	ROs shall approve or reject citizen registrations. Approval triggers the cryptographic handshake (activation challenge) sent to the citizen's device.	High
FR-A05	Birth and Death Record Approval	ROs shall review and approve or reject birth and death records submitted by health workers before certificates are finalized.	High

3.3.2 Registrar Functions

ID	Requirement	Description	Priority
FR-A06	Health Worker Enrollment	Registrars shall register healthcare professionals into the system, assigning credentials for Hospital App access.	High
FR-A07	Registration Officer Enrollment	Registrars shall enroll Registration Officers and assign them appropriate access permissions on the Administrative Dashboard.	High
FR-A08	Third-Party Vetting	Registrars shall review and approve or reject third-party institution applications for access to the GSB. Approved institutions receive OIDC client credentials.	High

ID	Requirement	Description	Priority
FR-A09	Integration Monitoring	Registrars shall be able to monitor the GSB integration of approved third-party institutions for compliance and misuse detection.	Medium

3.3.3 Data Analyst / Supervisor Functions

ID	Requirement	Description	Priority
FR-A10	National Statistics Dashboard	Supervisors shall access a real-time dashboard displaying national registration trends, including total enrollments, birth/death rates, and regional distribution.	High
FR-A11	Immutable Audit Trail	Supervisors shall access a complete, immutable audit log of all system actions including DIN queries, API key rotations, approvals, and rejections, with timestamps and actor information.	High
FR-A12	Anomaly Detection	The system shall provide tools for Supervisors to identify and flag abnormal activity patterns in the audit trail.	Medium

3.4 Government Service Bus (GSB) Integration

ID	Requirement	Description	Priority
FR-G01	Third-Party API Authorization	The GSB shall manage OIDC client credentials for approved third-party institutions, ensuring they comply with the Zambia Data Protection Act before accessing citizen data.	High

ID	Requirement	Description	Priority
FR-G02	KYC Data Routing	The GSB shall route KYC requests from third-party institutions to the Citizen App for consent, and return only consented data fields.	High
FR-G03	Mobile Money Integration	The GSB shall support integration with MTN MoMo, Airtel Money, and Zamtel Money for transaction authorization.	High
FR-G04	ISO Messaging Support	The GSB shall support ISO 8583 and ISO 20022 messaging standards for financial transaction orchestration.	Medium
FR-G05	Transaction Orchestration	The GSB shall manage the lifecycle of financial transactions from initiation through confirmation, notifying both the citizen and the requesting institution.	High
FR-G06	Legacy System Integration	The GSB shall provide integration with Zambia's existing National Registration system to enable cross-referencing during citizen enrollment.	High

4. Non-Functional Requirements

4.1 Performance Requirements

ID	Requirement	Description	Priority
NFR-P01	API Response Time	All REST API endpoints shall respond within 2 seconds under normal load conditions (up to 1,000 concurrent users).	High
NFR-P02	Mobile App Load Time	The Citizen App shall display the digital ID within 3 seconds of a verification request, including offline scenarios.	High
NFR-P03	Cryptographic Operation Speed	QR code generation (signing) shall complete within 1 second on mid-range Android devices.	Medium
NFR-P04	Scalability	The system architecture shall support national-scale deployment, handling up to 19 million registered citizens.	High
NFR-P05	Offline Sync	The Hospital App shall successfully sync all queued offline records within 60 seconds of reconnection.	Medium

4.2 Security Requirements

All components of the ZDID System must adhere to the following security standards:

Requirement	Standard / Specification	Applies To
Authentication	OAuth 2.0 / OpenID Connect	All user-facing applications
Encryption at Rest	AES-256	Citizen App local storage, all databases

Requirement	Standard / Specification	Applies To
Encryption in Transit	TLS 1.3	All network communications
Digital Signatures	RSA-2048 or ECDSA P-256	Citizen ID binding, QR codes, certificates
Hashing	SHA-256 or SHA-3	All stored credentials and sensitive data
Key Management	HSM-backed or cloud KMS	All cryptographic key operations
Audit Logging	Immutable trail with timestamps	All system actions across all components
Data Privacy	Zambia Data Protection Act	All citizen personal data
Biometric Standards	ISO/IEC 19794 series	All biometric capture and storage
Anti-Deepfake	Liveness detection during biometric capture	Registration Officer enrollment stations

4.3 Usability Requirements

ID	Requirement	Description	Priority
NFR-U01	Multi-language Support	All three applications shall support English, Bemba, Nyanja, Tonga, Lozi, Kaonde, and Lunda. Language shall be selectable at any time by the user.	High
NFR-U02	Accessibility	The Citizen App shall comply with WCAG 2.1 Level AA accessibility standards to accommodate users with visual or motor impairments.	Medium

ID	Requirement	Description	Priority
NFR-U03	Low-Literacy Design	The Citizen App shall use iconography and visual cues in addition to text, to support users with lower literacy levels.	Medium
NFR-U04	Print Support	Digital certificates (birth, death, ID) shall have a print-friendly format for citizens without smartphones.	High

4.4 Reliability and Availability

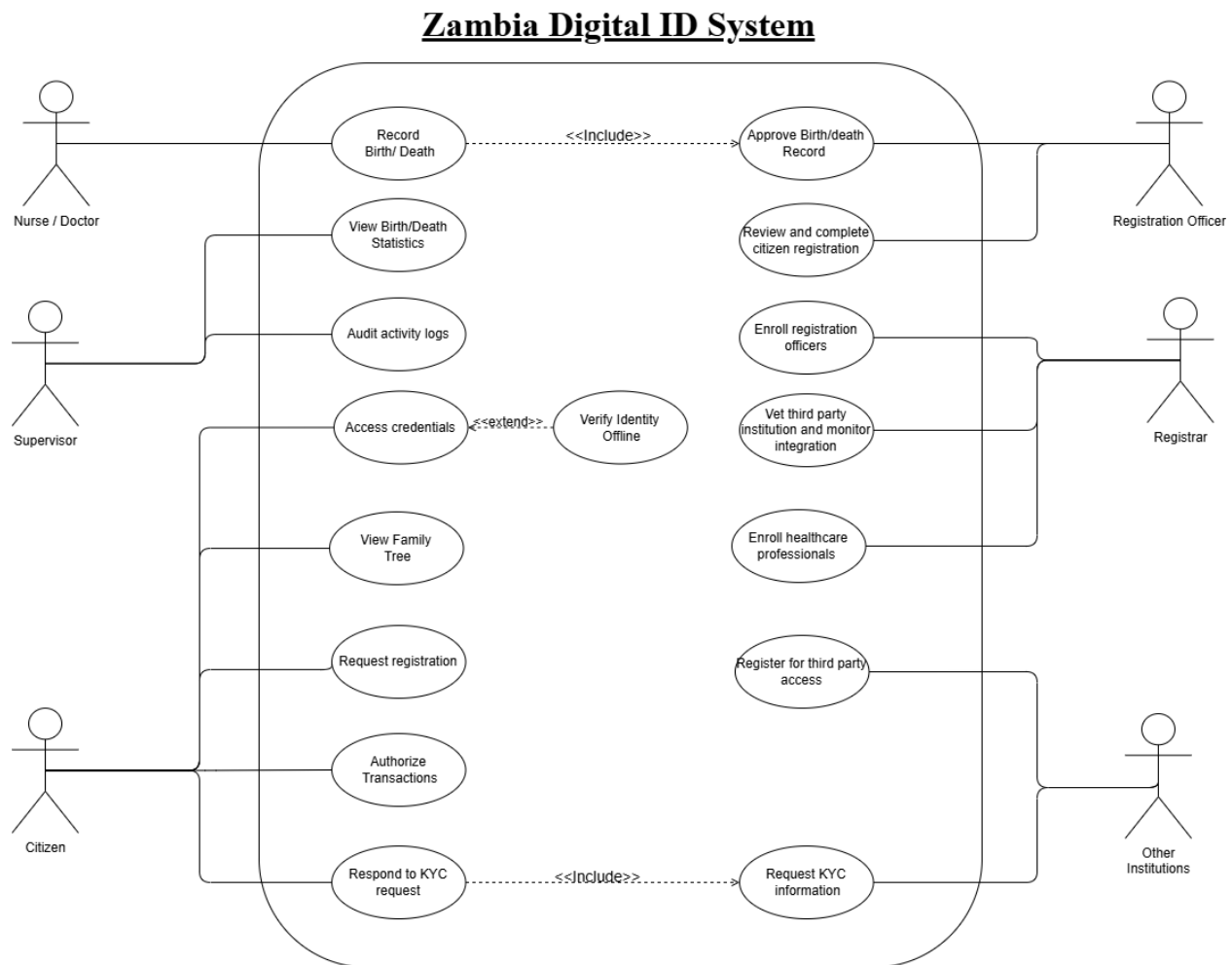
- The system shall target 99.5% uptime for core identity services
- The Citizen App shall remain fully functional for identity display and verification in offline mode
- The Hospital App shall queue offline records without data loss for up to 72 hours without connectivity
- All data modifications shall be logged to the immutable audit trail before being committed

5. System Design Diagrams

This section presents the UML design diagrams produced during the design phase of the Zambia Digital ID System. These diagrams illustrate the system's architecture, actor interactions, process flows, and component communication.

5.1 Use Case Diagram

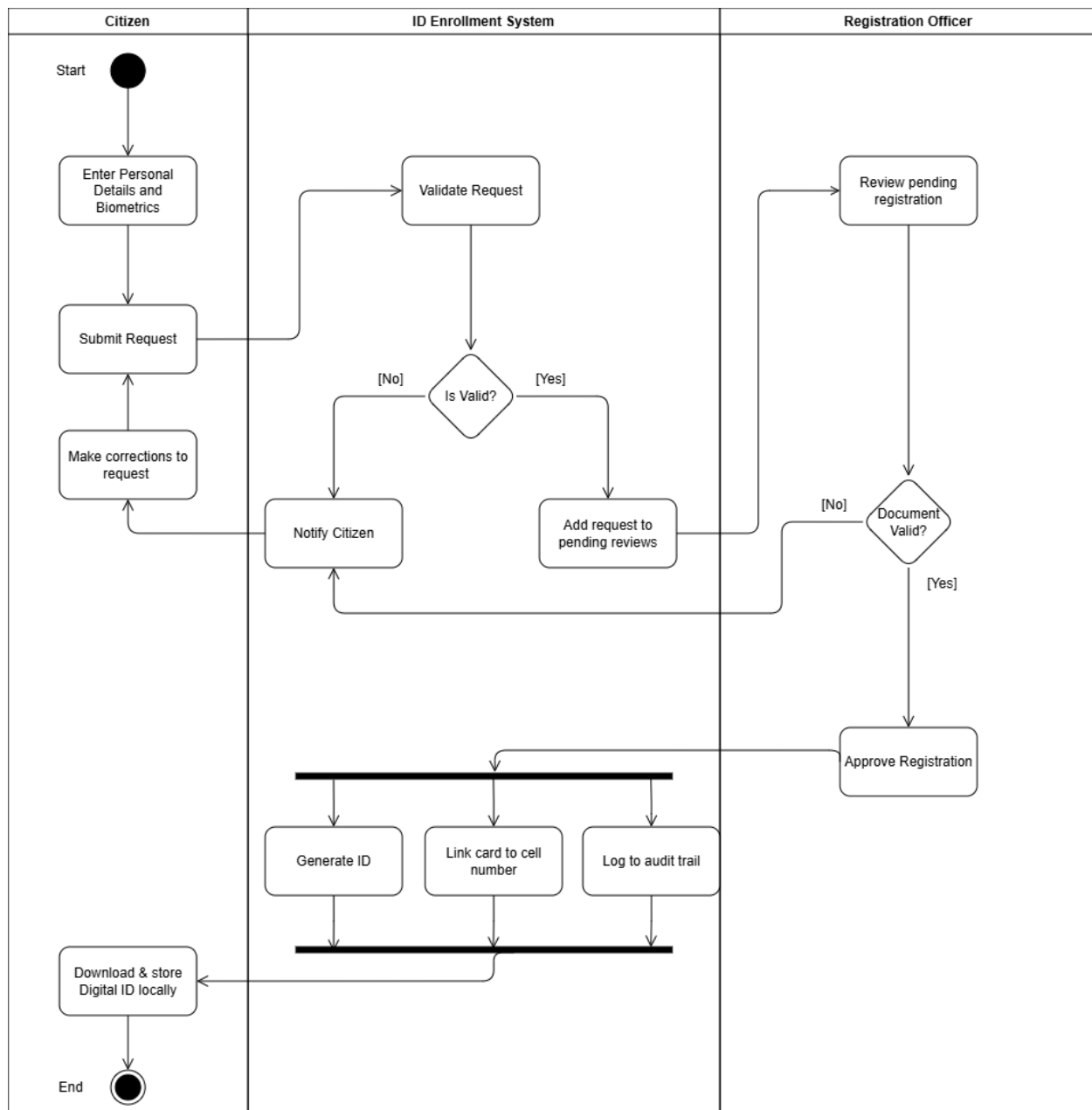
The use case diagram below illustrates the primary interactions between all system actors and the functionalities of the ZDID System. Actors include: Citizen, Healthcare Professional (Nurse/Doctor), Registration Officer, Registrar, Supervisor/Data Analyst, and Other Institutions (third-party).



5.2 Activity Diagrams

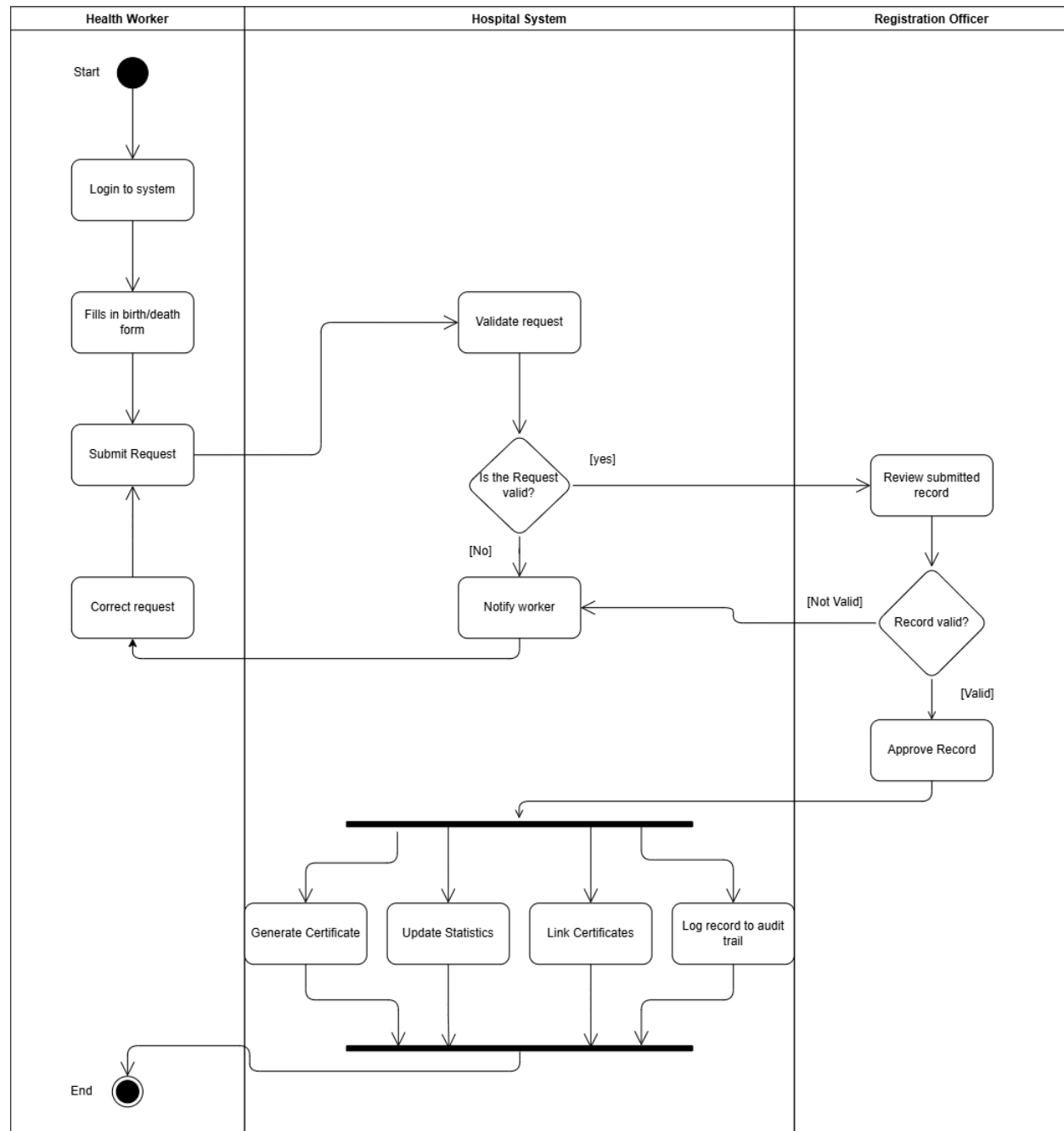
5.2.1 Citizen Registration Activity Diagram

This activity diagram shows the end-to-end process flow for citizen digital ID enrollment, from initial personal data entry through biometric capture and digital ID download.



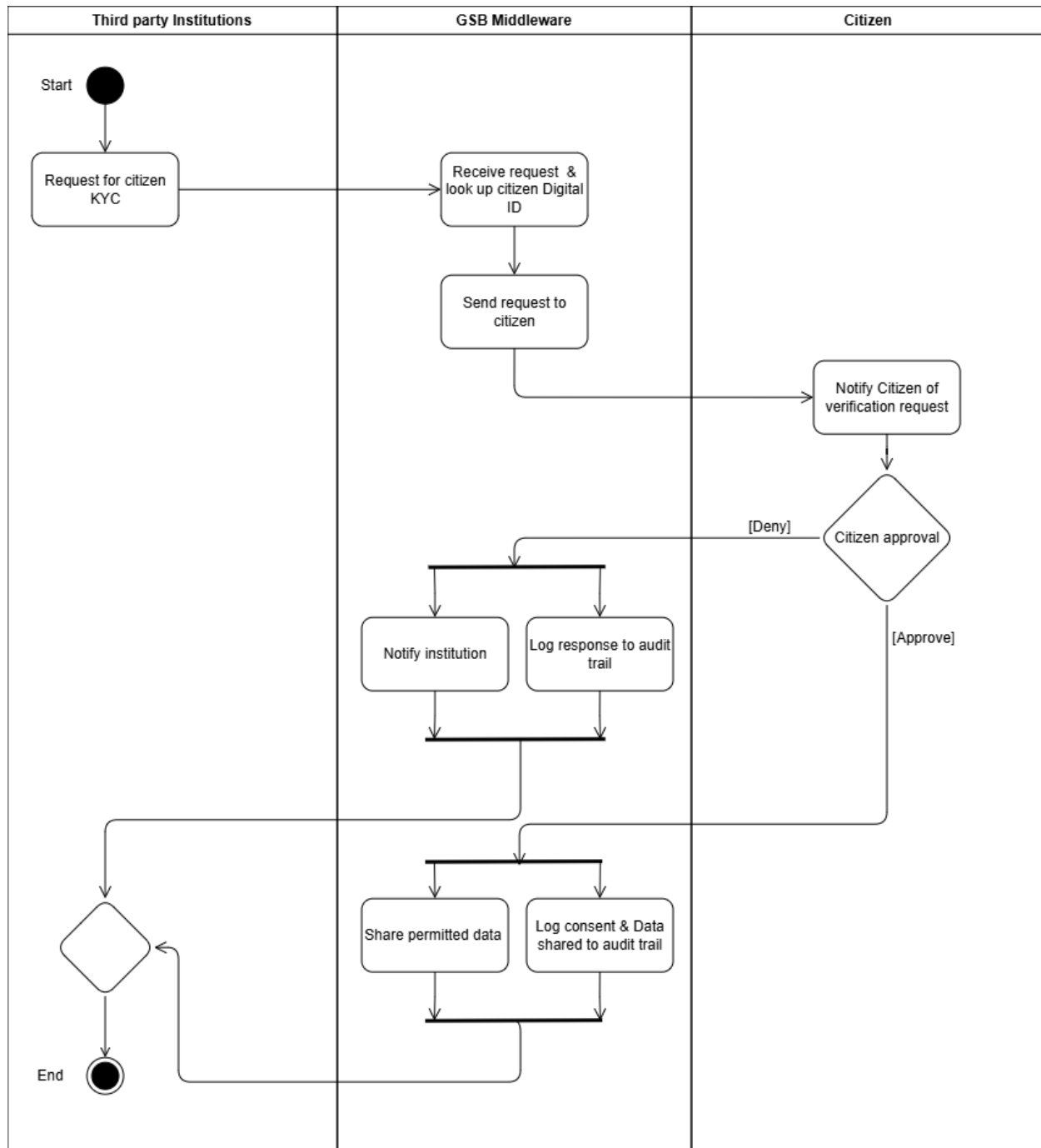
5.2.2 Birth and Death Recording Activity Diagram

This diagram illustrates the process for health workers recording vital events (births and deaths) through the Hospital App, including validation, Registration Officer approval, and parallel certificate generation.



5.2.3 Third-Party KYC Verification Activity Diagram

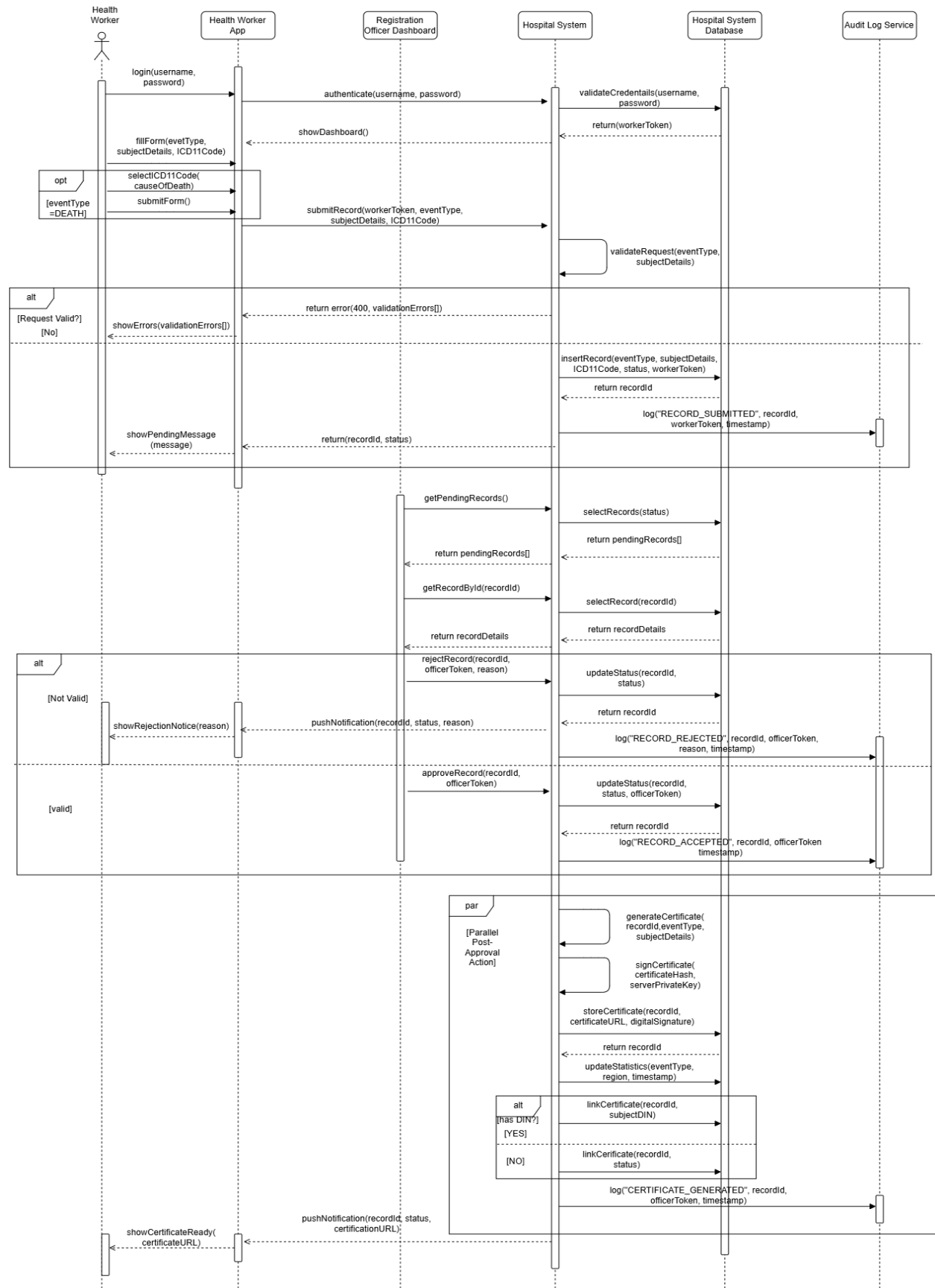
This diagram depicts the consent-driven KYC data sharing process between third-party institutions, the GSB Middleware, and the citizen, showing both approval and denial paths.



5.3 Sequence Diagrams

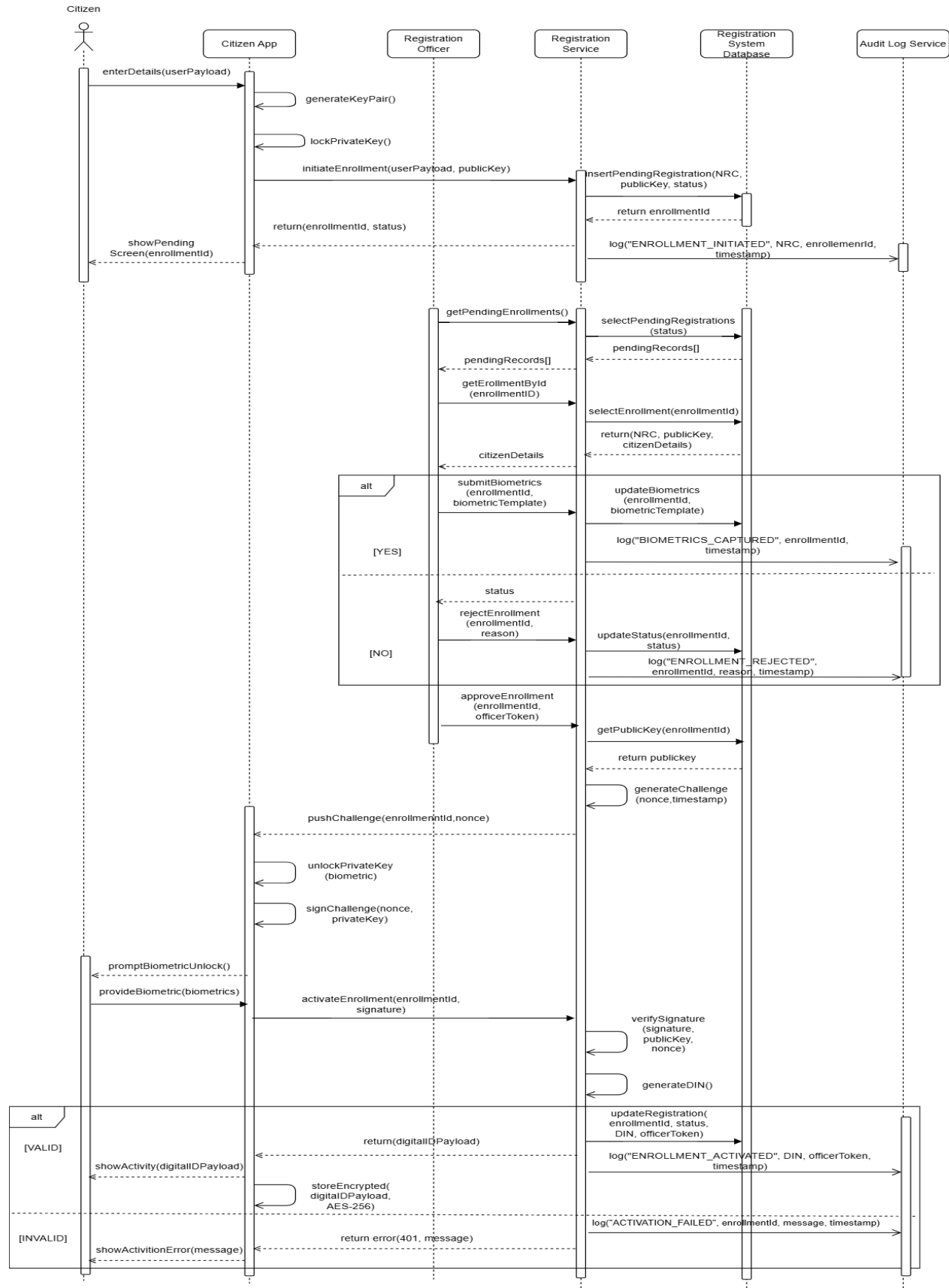
5.3.1 Birth & Death Recording Sequence Diagram

This sequence diagram details the message flow for recording a birth or death into the system, showing the interaction between the health worker, Registration Officer and backend services.



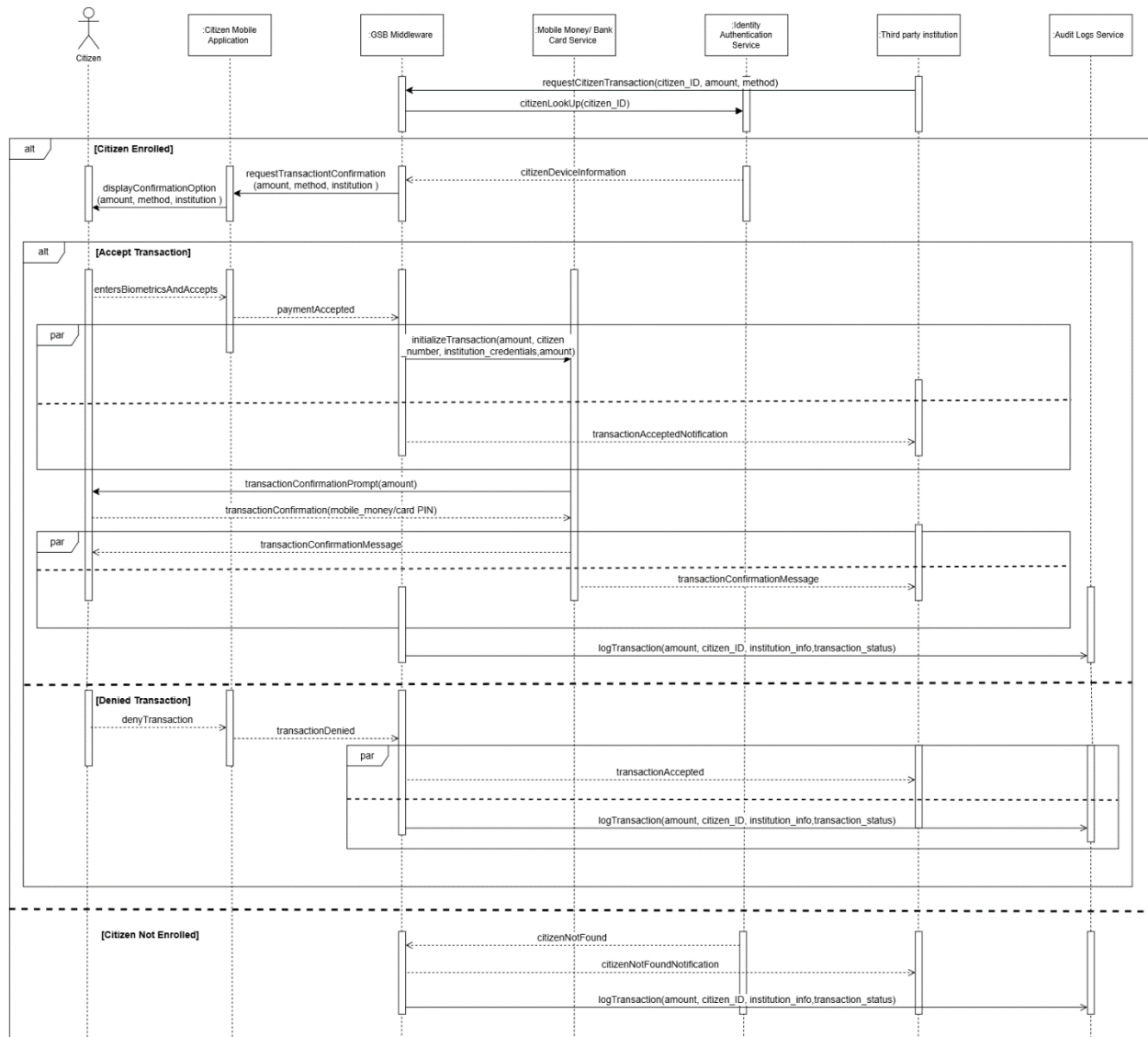
5.3.2 Citizen Registration Sequence Diagram

This diagram shows the complete sequence of interactions during citizen enrollment, from the initial cryptographic key generation on the citizen's device through physical verification by the Registration Officer and final DIN activation.



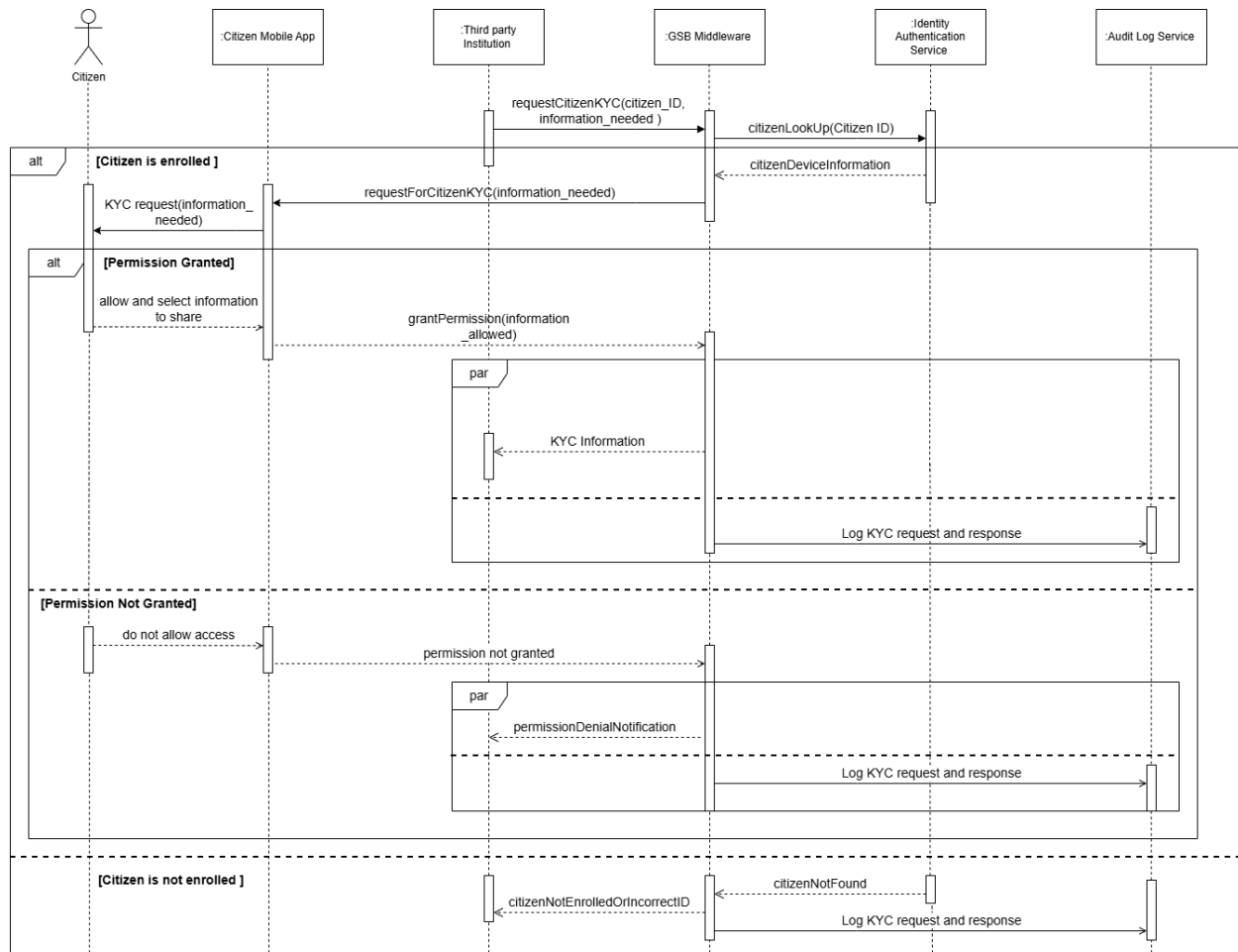
5.3.3 GSB Payment Orchestration Sequence Diagram

This sequence diagram illustrates the transaction authorization flow, showing how a third-party institution initiates a payment request, how the GSB verifies the citizen's identity, how the citizen authorizes via biometrics, and how the result is notified to all parties.



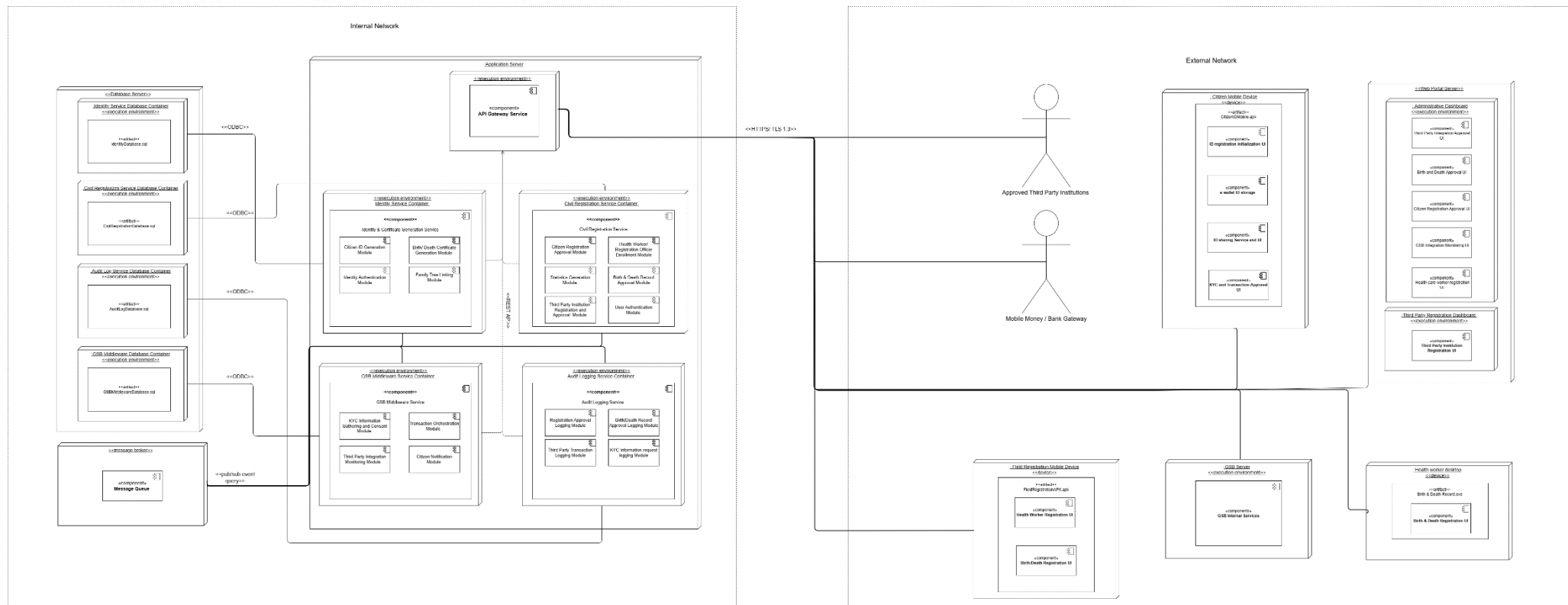
5.3.4 Third-Party KYC Access Sequence Diagram

This diagram details the message-level interactions for KYC data access, including citizen lookup, consent notification, permission granting or denial, and parallel logging to the audit trail.



5.4 Architecture Deployment Diagram

The deployment diagram below illustrates the physical and logical distribution of the ZDID System components across the internal network (application servers, service containers, and databases) and the external network (citizen mobile devices, third-party institutions, mobile money gateways, and the health worker mobile app).



6. System Architecture

6.1 Architecture Overview

The ZDID System follows a microservices-based, cloud-native architecture. All services are containerized using Docker and orchestrated via Kubernetes. An API Gateway serves as the single-entry point for all client requests, routing traffic to the appropriate microservice.

6.2 Core Services

- **Identity Authentication Service:** Manages citizen DIN generation, cryptographic key pair binding, and digital ID activation.
- **Civil Registration Service:** Handles all birth and death record processing, certificate generation, family tree linking, and statistics.
- **GSB Middleware Service:** Orchestrates all communication between internal services and external third-party institutions, including KYC routing and payment orchestration.
- **Audit Logging Service:** Maintains an immutable, timestamped log of all system events across all services.
- **API Gateway Service:** The single external-facing endpoint enforcing authentication, rate limiting, and request routing.

6.3 Recommended Technology Stack

Layer	Technologies
Backend	Python (FastAPI/Django)
Frontend (Dashboard)	React

Layer	Technologies
Mobile (Citizen & Hospital)	React Native (iOS and Android)
Database	PostgreSQL (primary), Redis (caching)
Message Queue	RabbitMQ
API Gateway	Kong
Containerization	Docker, Kubernetes
CI/CD	GitHub Actions

7. Data Requirements

7.1 Data Standards

- Personal Data: ISO/IEC 5218 for gender encoding; ISO 3166-1 for country codes
- Date and Time: ISO 8601 format throughout all records
- Biometrics: ISO/IEC 19794 series for fingerprint and facial data storage and exchange
- ID Card Dimensions: ISO/IEC 7810 for any physical card specifications
- Travel Documents: ICAO Doc 9303 for passport-related data fields

7.2 Key Data Entities

The system manages the following primary data entities:

- Citizen Record: DIN, NRC number, full name, date of birth, biometric templates, device public key, registration status, family linkages

- Birth Record: Child DIN, mother DIN, date/time/location of birth, health worker ID, certificate ID, ICD-11 codes (if applicable), approval status
- Death Record: Deceased DIN, date/time/location of death, cause of death (ICD-11), health worker ID, certificate ID, approval status
- Audit Log Entry: Timestamp, actor ID, action type, target resource, IP address, outcome, digital signature of log entry
- Third-Party Institution: Institution name, registration number, OIDC credentials, permitted data scope, vetting status, integration logs
- Transaction Record: Transaction ID, citizen DIN, institution ID, amount, method, status, timestamps

7.3 Data Privacy and Retention

All citizen personal data is subject to the Zambia Data Protection Act. Key privacy principles implemented include:

- Data Minimization: Only data strictly necessary for the purpose is collected and shared
- Consent-Based Sharing: No citizen data is shared with third parties without explicit, informed consent recorded in the audit trail
- Right to Access: Citizens can view all data held about them through the Citizen App
- Immutable Audit Trail: All data access events are permanently recorded and cannot be modified or deleted
- Secure Disposal: Retired identities (upon death) are cryptographically invalidated while historical records are preserved per legal requirements

8. Key User Stories

The following user stories capture the primary value propositions of the ZDID System from each stakeholder's perspective:

8.1 Citizen Stories

- Hybrid Enrollment: As a Zambian citizen, I want to register on my mobile app by entering my NRC details, set up biometric and generating a cryptographic key pair, so that I can get my digital ID approved.
- Secure Login: As a registered citizen, I want to use my phone's biometrics to sign a dynamic challenge when logging into my bank, so that I can prove my identity without needing to remember a password or worrying about phishing.

8.2 Healthcare Professional Stories

- Vital Event Story: As a nurse at a local clinic, I want to digitally record a newborn's birth and link it to the mother's 12-character DIN, so that the government can automatically trigger the issuance of a digital birth certificate.
- Mortality Reporting: As a health worker, I want to record a death using ICD-11 classification codes, so that national health statistics are accurate and the deceased's identity is securely retired in the registry.

8.3 Administrative Stories

- Identity Verification: As a Registration Officer, I want to verify a citizen's documents against the legacy National Registration system via the GSB, so that I can confidently approve the binding of their physical identity to their new digital device.
- Administrative Oversight: As a Registrar, I want to manage the access levels of health workers and field officers, so that only authorized personnel can enter or approve sensitive identity data.

8.4 Security and Governance Stories

- **API Authorization:** As a Third-Party Manager, I want to vet and authorize a bank's OIDC client credentials, so that I can ensure they comply with the Zambia Data Protection Act before they can access citizen data.
- **Audit Integrity:** As a system auditor, I want to view an immutable audit trail of every DIN query and API key rotation, so that I can detect and investigate any unauthorized attempts to access the national registry.

9. Appendix

9.1 Document Revision History

Version	Date	Description	Author
1.0	April 2026	Initial SRS - Design Phase Submission	Stack Overlords

9.2 References

1. ZAMREN Student Hackathon 2026 - Project Brief: Zambia Digital ID T
2. Zambia Data Protection Act - Data Privacy Regulations
3. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
4. OAuth 2.0 Authorization Framework - RFC 6749
5. OpenID Connect Core 1.0 Specification
6. ICD-11 International Classification of Diseases - World Health Organization
7. ISO/IEC 19794 Biometric Data Interchange Formats
8. ICAO Doc 9303 - Machine Readable Travel Documents
9. ISO/IEC 7810 - Identification Cards Physical Characteristics

End of Document

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